

Lab 17: Ecology — Bioaccumulation & Biomagnification of DDT

BIOL-8

Name: _____ Date: _____

Learning Objectives

- Explain **bioaccumulation** (buildup of a contaminant within an individual) and **biomagnification** (increasing concentration at higher trophic levels).
- Track **DDT bead** counts through a model food chain: producers → stickleback → cutthroat trout → osprey.
- Interpret bead data to describe how **DDT** moves and concentrates in a simplified aquatic web.
- Connect **DDT** to documented effects on **top predators** (e.g., eggshell thinning in birds) and to **indirect** effects on other populations when a predator declines.

Purpose

Bioaccumulation and biomagnification of DDT: You will observe how a contaminant accumulates in organisms and becomes more concentrated at higher levels of a simple food web, using beads to model producers, fish, and a fish-eating bird.

Materials

- **Shaker cup (plastic)** — holds the producer population
- **9 small cups** — stickleback fish
- **3 medium cups** — cutthroat trout
- **1 large cup** — osprey
- **20 beads of one color** — producer biomass
- **10 beads of a second color** — DDT (pesticide)

- **Scratch paper** (optional) — tally DDT in each small/medium/large cup before you copy **summary** numbers into the data table

Total: 30 beads in the shaker at the start, with **DDT making up 1/3** of the starting mixture (10 of 30 beads).

Background

This lab explores **bioaccumulation** (buildup of contaminants within an individual organism) and **biomagnification** (increasing concentration at higher trophic levels), using **DDT** as the case study. DDT was inexpensive to manufacture and persistent in the environment. It enters aquatic systems by binding to plankton, accumulates in their cells, passes to fish that eat the plankton, and then to birds that eat the fish. In birds, DDT disrupts **eggshell formation**, producing brittle shells that break before hatching.

Procedure

Setup

1. Place all **30 beads** (20 producer + 10 DDT) into the **shaker cup**.
2. In the data table, record the **DDT-to-total ratio** in the shaker ($10/30 = 1/3$). Shake thoroughly to mix.

Trophic level 1 → 2 (producers → stickleback)

1. Remove **3 beads** from the shaker and place them into **each** of the **9** stickleback cups (27 beads moved).
2. Tally the **number of DDT beads in each of the 9** cups on **scratch paper** (or the margin) if you like. You do **not** need to list every cup on this handout. For **this trial**, compute the **average DDT per stickleback** = **(sum of DDT in all 9 cups) ÷ 9** and enter that **one number** in the **Data** table in the **column** for the trial you just ran (**Trial 1, 2, or 3**; see **Data** below).

Trophic level 2 → 3 (sickleback → cutthroat trout)

1. Each trout eats **2 sickleback** — empty **2 small cups** into **each** of the **3** medium (trout) cups.

2. Tally **DDT in each of the 3 trout cups** if needed, then for **this trial** record **average DDT per trout** = **(sum of DDT in the 3 trout cups) ÷ 3** in the **Data** table in the same trial **column** as in step 4.

Trophic level 3 → 4 (trout → osprey)

1. Each osprey eats **2 trout** — empty **2 medium cups** into the **large** (osprey) cup.
2. For **this trial**, record **total DDT in the osprey** (one **large** cup) in the **Data** table in the same trial **column**. (There is one osprey: “average DDT per osprey” is the **same** as that total.)

Trials and calculations

1. **Repeat** the full procedure (setup through osprey) for **Trials 2 and 3**. Each new trial, **replace** all 30 beads in the shaker, remix, and fill in the same rows for that trial’s column.
 2. For each row, you may also compute a **mean across the three trials** in the last column. Use the **Data** table as your only required record on this page.
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Data

Tally on **scratch paper** as needed. Enter **summary** values only: **one column per trial** (plus an optional **mean** column).

Averages for stickleback and trout use **(sum in cups) ÷ 9** and **(sum in 3 fish) ÷ 3**, not per-cup columns on the handout. **T1–T3** = trial columns; **Mean** = optional average after all three trials.

DDT summary — three trials (bead counts or averages)

Metric	T1	T2	T3	Mean
Initial — DDT / 30 (~1/3)				
Stickleback — avg DDT (÷9)				
Trout — avg DDT (÷3)				
Osprey — total DDT (large cup)				
Notes (optional)				

Analysis questions

Answer in **complete sentences**. Use your **data table** where a question asks for numbers or patterns.

Warm-up (introductory). Short prompts to connect the procedure to the questions below.

A. In this model, what did the **producer-colored** beads represent, and what did the **DDT-colored** beads represent?

B. Before you cite your table: in plain language, what is the difference between **bioaccumulation** (buildup in one organism) and **biomagnification** (change **up** the food chain) in this lab?

C. The handout asks for an **average** DDT per stickleback using all nine cups. Why is that average a better summary than writing down only **one** stickleback cup?

1. Using the **Data** table (stickleback, trout, and osprey for each **trial**), state how **average DDT per organism** (or **total** for the osprey) **compared** across trophic levels. If your instructor wants **extra** work, run the averages in the **last column** only after all three trials are done.

2. Construct a **food chain** illustrating the flow of energy through this model (from producers through osprey). Label trophic levels if asked.

3. What happened to the **amount of DDT** as you moved **up** the food chain?

4. Why is **DDT harmful** to the osprey?

5. Name **another organism** (besides osprey) you would expect to have **high DDT concentrations**, and explain why.

6. If the **osprey population declines** due to DDT, name **two other populations** that would be affected and describe **how**.